

Exploratory Data Analysis (EDA)

Using Pandas, Seaborn & Matplotlib

Introduction to EDA

- Exploratory Data Analysis (EDA) helps in understanding datasets
- Tools used: Pandas, Seaborn, Matplotlib
- Steps: Summary statistics, Visualization, Feature relationships

Step 1: Load Dataset

- Use Pandas to load CSV or Seaborn built-in datasets
- Example:

```
df = sns.load_dataset('titanic')
```
- Inspect with `df.head()`

Step 2: Summary Statistics

- `df.shape` – dataset dimensions
- `df.info()` – datatypes & missing values
- `df.describe()` – statistical summary
- `df.isnull().sum()` – missing values count

Step 3: Univariate Analysis

- Study single feature distribution
- Numerical: Histograms, KDE plots
- Categorical: Count plots
- Example: `sns.histplot(df['age'])`

Step 4: Bivariate Analysis

- Explore relationship between two features
- Scatter plots: `sns.scatterplot(x, y)`
- Box plots: `sns.boxplot(x, y)`
- Groupby with Pandas:
`df.groupby('class')['fare'].mean()`

Step 5: Correlation Analysis

- Correlation matrix using `df.corr()`
- Visualize with Heatmap:
`sns.heatmap(corr, annot=True)`
- Helps identify feature relationships

Step 6: Pairwise Relationships

- Use `sns.pairplot()` for small datasets
- Visualizes multiple feature relationships
- Example: `sns.pairplot(iris, hue='species')`

Step 7: Handling Missing Values & Outliers

- Missing values:
 - `df.dropna()`
 - `df.fillna(mean/median)`
- Outliers:
 - Detect using boxplots
 - Handle with capping or removal

EDA Summary Roadmap

1. Load & Inspect Data
 2. Univariate Analysis
 3. Bivariate Analysis
 4. Correlation Analysis
 5. Pairwise Relationships
 6. Handle Missing Values & Outliers
- ✓ Prepares data for Machine Learning